

Posture Correction and Shuttle Metrics Analysis for Badminton using Computer Vision

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Abstract—Automated analysis of badminton gameplay from raw video remains challenging due to rapid shuttle motion, occlusions, and the need to jointly model biomechanics and tactical context. This paper presents an end-to-end computer vision framework for posture correction and shuttle metrics analysis using monocular match footage. Using court homography estimation and multi-person pose tracking combined with physics-aware shuttlecock tracking through Kalman filtering techniques, the proposed system introduces an automated Impact Score algorithm, a heuristic metric derived from the product of normalized shuttlecock acceleration and angular inflection to localize hit events without manual annotation. The transformer-based multimodal classifier uses biomechanical/spatial data and shuttlecock motion information to classify shots. A confidence-gated tactical model uses the entire rally context to strengthen the confidence level for shot classification. The system provides personalized form correction by comparing user kinematics against professional biomechanical benchmarks, delivering actionable feedback (e.g., “Add Elbow Extension”) when deviations in joint angles are detected. Most existing techniques treat hit detection, shot classification, biomechanical assessment and strategic reasoning individually, frequently using manually annotated data or external sensors as input. An automated, fully integrated approach that automatically analyses all four components with raw footage in real-time is still a challenge. Validated on a diverse dataset of 8,232 stroke samples across 9 classes, combining professional match footage with amateur recordings to ensure cross-domain robustness, the proposed system was validated with accurate hit detection, stable trajectory reconstructions, and dependable shot classifications regardless of recording conditions, confirming its potential use in automated coaching for badminton using video data.

Index Terms—Computer vision, badminton analytics, pose estimation, shuttlecock tracking, transformer models, biomechanical analysis

I. INTRODUCTION

Badminton is a fast-paced sport requiring precise timing, technique, and shot selection. Many amateur and intermediate players lack access to professional coaching and therefore rely on subjective assessment or basic video review, which

provides limited insight into technique and tactical decision-making. Recent advances in computer vision and deep learning have enabled automated video analysis for sports, including badminton, through tasks such as shuttle tracking, pose estimation, and shot classification. However, most existing systems address these components independently and continue to rely on manual annotation, embedded sensors, or controlled recording conditions, limiting their applicability in real match scenarios. State-of-the-art shuttlecock tracking is represented by the TrackNet series (V1–V3), which utilizes heatmap-based convolutional neural networks and temporal modeling to detect the shuttlecock under motion blur. MonoTrack further improves single-view trajectory association and 3D reconstruction from monocular video. However, these approaches are primarily data-driven and often struggle with depth ambiguity and the non-parabolic trajectories of high-speed smashes. This paper presents a fully automated badminton video analysis framework that operates directly on raw match footage. The proposed system integrates shuttle tracking, hit event detection, biomechanical feature extraction, and transformer-based shot classification within a unified pipeline. By combining shuttle dynamics and player biomechanics, the framework distinguishes visually similar shots and associates them with the correct player. Furthermore, it quantifies amateur player performance relative to professional reference distributions, enabling scalable and practical coaching support using standard video recordings.

II. LITERATURE SURVEY

Recent research in badminton has increasingly applied computer vision and deep learning techniques to extract technical and tactical insights from match videos. Early efforts were restricted by limited datasets which constrained scalability and contextual understanding. To address this limitation, Wang *et al.* introduced *ShuttleSet*, a large-scale, human-annotated dataset containing stroke-level annotations for singles bad-

minton matches. This dataset enabled direct learning for shot classification, player movement analysis, and tactical modeling [1]. Extending this line of work, He *et al.* proposed *FineBadminton*, a multi-level dataset that captures stroke actions, tactical semantics and rally context, supporting fine-grained video understanding [8].

Due to high speed and small size of shuttlecock, and the fact that it is often surrounded by motion blur, a tracking method cannot accurately identify the position of the shuttlecock, or any object moving at high speeds. Chen *et al.* developed TrackNet, a deep learning-based framework that significantly improved the efficiency of shuttlecock tracking in sports videos. This was achieved by using a set of temporal heatmaps, with one heatmap generated for each frame over time, which allowed the distribution of the shuttlecock to be analyzed across frames and its motion to be observed over a given time period. Through continuously updated heatmaps, it became easier to estimate the shuttlecock's position even under challenging conditions such as motion blur caused by camera movement, sudden changes in direction, or partial occlusion by obstacles. With the previous version of TrackNet (v1), consecutive images were stacked together to create a probability heatmap of where the shuttle was located over time. Both TrackNet V2 and V3 added a deeper convolutional backbone and improved the temporal aggregation to allow for greater robustness with respect to sudden direction changes and rapid motion. In addition, these advancements reduced the number of false detections and stabilized the tracking of shuttles through difficult rally sequences [2]. There have also been attempts to estimate and predict the path of the shuttlecock using machine learning techniques, such as Yang's use of a neural network to create a more precise estimate of the shuttlecock trajectory [10]. Other approaches, such as particle filters and physics-based models, were also developed to further improve the tracking of shuttle flight prediction accuracy in noisy video environments [12].

Several studies have focused on using lightweight object detection models for badminton analysis in real-time. Cheng and Kim used the YOLOv5-based architectures to detect shuttlecocks, players and court areas from the broadcast video in real-time, demonstrating the feasibility of real-time implementation [3]. However, the object detection pipeline was incapable of identifying very similar images. In contrast, the use of pose-based representations of shots showed promise in improving the identification of shots. Salim *et al.* proved that a skeletal keypoint-based model was a better option than an object-based model in the classification of badminton shots, as pose feature provided a better ability to observe subtle differences among player's preparations and executions [5].

Pose estimation has also enabled the development of automated coaching and posture correction systems. Chai *et al.* developed DL-Shuttle, which utilized deep learning to provide badminton technique feedback via pose estimation, enabling access to these services for beginner players who do not use wearable sensors [6]. Nasution *et al.* further demonstrated that machine learning-based training models could improve fun-

damental badminton skills through video-generated feedback [4]. Despite being effective for basic coaching, these systems primarily focus on individual postures and did not take into account shot-specific biomechanics.

Beyond vision-only approaches, wearable sensor systems were used to detect strokes. Kim and Park applied deep learning with accelerometer and gyroscope data to identify badminton strokes, achieving high detection accuracy [9]. Although effective, these methods relied on additional hardware and did not utilize the spatial and tactical data available in videos. Hybrid systems that combined video data with learning-based reasoning were also proposed. Toshniwal *et al.* developed an AI coaching framework to evaluate badminton performance by combining pose estimation, shuttle tracking, and object recognition [11]. Similarly, RallyNet has developed methods of modeling how players behave and what sequence of shots they hit using both experiential and stochastic motion modeling for offline imitation learning [7].

Subsequent work focused on advanced methods for trajectory reconstruction and tactical analysis. Liu and Wang proposed a framework called MonoTrack, a physics-informed optimization framework, used to reconstruct 3D shuttle trajectories from 2D monocular badminton video sequences. As opposed to purely detection-based approaches, MonoTrack leveraged aerodynamic and trajectory fitting constraints in order to enhance both the spatial consistency and depth estimation derived from the 2D input. Therefore, MonoTrack allowed for improved visualization of the shuttle's flight trajectory and provided an alternative means for evaluating a player's tactical movement without the requirement of multiple camera views. [14]. Other methods involving player position estimation based on multi-person pose estimation and court calibration, were also applied to tactical analysis [15]. Additionally, Yue *et al.* developed a method to segment rallies using pose-based temporal features to detect ready-to-serve actions [16]. Comparison techniques based on Part Affinity Fields (PAFs) were developed and applied to objectively evaluate players' postures and movements [13].

Table I highlights the key differences between existing badminton video analysis methods from the proposed framework. Overall, current research on individual components of badminton analysis, such as shuttle tracking, pose-based shot classification, trajectory prediction, and virtual coaching, has come a long way; however, most existing methodologies treat these different areas individually, frequently using manually annotated data or external sensors as input. An automated, fully integrated approach that automatically analyses all four components (i.e., hit detection, shot classification, biomechanical assessment and strategic reasoning) with just raw footage in real-time remains an unmet need. This paper attempts to address this issue.

III. METHODOLOGY

The proposed system utilizes computer vision, biomechanical pose, shuttlecock trajectory modeling, and temporal learning algorithms in a single unified system to automate recog-

TABLE I
COMPARATIVE SUMMARY OF PRIOR WORK AND PROPOSED SYSTEM

Aspect	Prior Work	Proposed System
Shuttle Tracking	TrackNet [2] and MonoTrack [14] achieve accurate localization but do not perform hit attribution.	Tracking combined with physics-based hit detection using a kinematic impact score.
Hit Detection	Datasets such as ShuttleSet [1] rely on manual hit annotations; wearable-based systems [9] require sensors.	Fully automated hit detection from raw monocular video using jerk-spike analysis.
Shot Classification	Pose-based models [5] and trajectory-based approaches [10] classify strokes independently, causing ambiguity in similar shots.	Biomechanical features fused with rally-level tactical context via Transformer-based late fusion.
Feedback	DL-Shuttle [6] provides posture feedback without shot-specific benchmarking.	Shot-specific biomechanical benchmarking against professional reference.
Integration	Hybrid systems such as AI Coach [11] and RallyNet [7] address components separately without a unified end-to-end pipeline.	Unified end-to-end system integrating tracking, hit detection, classification, and feedback.

dition of individual shots and improve player postures from match video footage. An overview of the complete pipeline is shown in Fig 1.

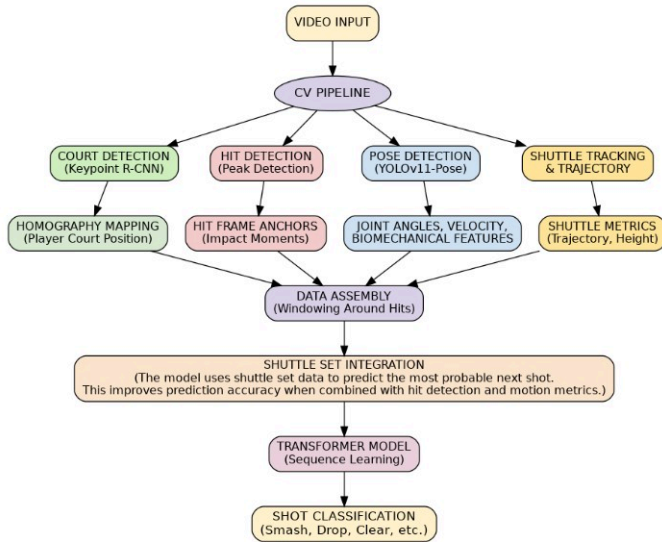


Fig. 1. End-to-end methodology pipeline for automated badminton shot recognition and posture correction.

A. System Overview

The proposed solution manages the unedited badminton match video through four distinct stages: visual perception and spatial normalization, event detection and temporal segmentation, multi-modal feature modeling and shot classification, and biomechanical benchmarking and feedback generation. Each of the four stages of processing occurs simultaneously

but is synchronized to the hit events detected in the initial phase of processing, thus ensuring effectiveness with respect to variations in camera angles and lighting conditions; as well as the varying skills of players.

B. Court Detection and Homography Normalization

The keypoint based system detects the badminton court and is then used to determine the intersection points on the court. These intersection points are used to compute a planar homography, which provides a uniform map of the camera's view with respect to a top view of the badminton court in a canonical coordinate space. Therefore, by normalizing the locations of all players, shuttle flight paths, etc., to a common spatial reference frame, we reduce the impact of the positioning of the camera and perspective distortion on how we view the spatial relationships between players and shuttle flight paths. To handle varying camera viewpoints and recording conditions, a 4-point homography normalization is applied so that all player and shuttle coordinates are projected into a standardized court coordinate system. This ensures that the downstream classifier operates on geometrically normalized features rather than raw pixel positions, thereby improving robustness across broadcast and amateur footage.

C. Pose Estimation and Biomechanical Signal Extraction

A YOLO-based real-time pose estimation network is used to capture player posture by extracting two-dimensional (2D) skeletal keypoints from each frame of video. Using the 2D skeletal keypoints, we compute three-dimensional (3D) joint angles, joint angular velocities, and higher-order kinematic derivatives of the player's upper extremities that are important to the biomechanics of badminton strokes, such as shoulder rotation, elbow extension, and wrist extension. All joint angles are derived from projected 2D keypoints using vector geometry (dot-product formulation) and therefore represent planar approximations rather than full 3D anatomical reconstructions. Mechanisms to smooth the output data are implemented to reduce the effects of noise caused by motion blur and partial occlusion of the player's body when using a video-based analysis.

D. Shuttlecock Tracking and Trajectory Modeling

The shuttlecock trajectory is determined and monitored through a combination of learning-based detection and a Kalman filtering model. These two techniques allow for confident estimation of a shuttlecock's position and velocity during ultra-high-speed motion and instances when there may not be any detection of the shuttlecock at all. Rather than explicitly solving aerodynamic drag equations (which vary with shuttle orientation), the system employs Savitzky-Golay filtering (window = 5, polyorder = 2) to smooth positional data while preserving high-frequency kinematic inflections associated with impact events. This derivative-based approach ensures stable trajectory reconstruction even when the shuttle exhibits non-parabolic motion due to aerodynamic drag during smashes. By reconstructing the trajectory of the shuttlecock we

can derive information regarding its flight characteristics (i.e., maximum speed, maximum height, direction diversity, etc.) giving us valuable cues concerning the player's intended shot and the quality with which that shot was executed.

E. Hit Event Detection and Temporal Segmentation

Kinematic discontinuities identified in shuttle and player pose signals have been utilized to identify hit events temporally/localized. The hypothesis behind the method used to detect hit events is based on a *jerk spike*, whereby there will be sudden spikes occurring in higher orders of motion derivative that correspond to the moment of shuttle-racket contact. We formally define the hit event using a Kinematic Impact Score (St) calculated at frame t. Raw shuttle coordinates are first smoothed using a Savitzky–Golay filter (window = 5, polyorder = 2) to ensure derivative stability. The score is computed as shown in (1), the product of normalized acceleration magnitude and normalized angular inflection:

$$S_t = \text{Norm}(\|a_t\|) \times \text{Norm}(\Delta\theta_t) \quad (1)$$

where a_t represents the second-order derivative (acceleration magnitude) of shuttle position, and $\Delta\theta_t$ denotes the frame-to-frame directional change of the velocity vector. Candidate hit frames are identified as local maxima exceeding a prominence threshold of 0.08, with a minimum temporal separation of 0.25 seconds (approximately 8 frames at 30 FPS) enforced to prevent double-counting strokes. After identifying the frames at which hits occur, we can extract fixed-length temporal windows both before and after the impact. It is these windows that represent the essential components of all subsequent shot classification and biomechanical evaluation.

F. Multimodal Feature Representation

A multimodal feature vector is formed for each temporal window from components of pose-derived biomechanical data, statistical data about the shuttle trajectory, and homography-normalized coordinates representing the court in 3D space. The features are all normalized and sequence-windowed in order to preserve temporal coherence and promote the ability to learn from long-range dependencies. Biomechanical pose features and shuttle motion trajectories are first processed through independent encoding branches to extract high-level temporal embeddings. These embeddings are then integrated using a Late Fusion strategy within a Transformer architecture via cross-attention. This attention-based fusion enables the model to dynamically correlate specific postural movements (e.g., wrist snap or elbow extension) with shuttlecock impact dynamics, while preserving the distinct temporal characteristics of the visual and kinematic data streams.

G. Shot Classification via Confidence-Gated Neural Fusion

A two-tiered approach comprising of a biomechanical classifier and a tactical sequence predictor is used to classify shot recognitions. The biomechanical classifier focuses on posture-orientated movement patterns, while the tactical sequence predictor uses a Transformer Encoder to include temporary

dependencies at the rally level. Both models provide prediction outputs that are fused through a confidence-gated fusion mechanism, which combines both outputs according to their level of confidence in order to increase robustness against visually similar shots and uncertain movement patterns. The fusion mechanism is governed by a deterministic gate based on the primary classifier's softmax confidence (C_A), with a tuned threshold $\tau = 0.85$. If $C_A \geq \tau$, the system relies solely on the biomechanical classifier's prediction. Otherwise, a weighted fusion as shown in (2) is triggered such that:

$$P_{final} = 0.6 P_{Biomech} + 0.4 P_{Tactical}. \quad (2)$$

The weighting parameter $w = 0.6$ was selected empirically via grid search on the validation set to prioritize visual biomechanical evidence while still incorporating rally-level tactical context.

H. Biomechanical Benchmarking and Feedback Generation

The extracted biomechanical data is referenced against a set of Biomechanical Reference Metrics for Coaches. A "Professional Baseline" is established by computing the mean (μ) and standard deviation (σ) of key joint angles (e.g., elbow extension, shoulder rotation) from the ShuttleSet professional dataset. These joint angles are derived from projected 2D keypoints using vector geometry and represent planar biomechanical approximations. By analyzing the biomechanical data against the Reference Metrics, a coach will be able to identify biomechanical deviations from the Reference Metrics for a specific type of shot (i.e., forehand multiple angles) and determine the degree of difference of that biomechanical deviation from the reference. Deviation is quantified using a Z-score formulation, and feedback is triggered when a user's joint angle deviates beyond 1σ from the professional mean (corrective feedback), while deviations exceeding 2σ in the positive direction are interpreted as strong biomechanical extension and may trigger positive reinforcement. A coach can then provide the feedback for each biomechanical parameter along with a graphical representation of where the shot was hit on the training surface for specific training of that biomechanical parameter.

IV. RESULTS AND ANALYSIS

The reliability of hit detection, classification accuracy of shot type, consistency of biomechanical feedback, and impact of confidence-gated fusion on shot classification accuracy were evaluated. The experiments utilised a composite dataset consisting of professional television footage, curated match videos, and amateur recordings taken from different vantage points. The dataset comprises 8,232 stroke samples across nine shot classes, combining 50 professional women's singles matches from ShuttleSet with amateur recordings from the VideoBadminton repository. To prevent data leakage, a Leave-One-Match-Out split strategy was employed, ensuring that entire matches or video files were assigned exclusively to training, validation, or test sets. This guarantees evaluation on unseen players and court environments.

To detect hit events, kinematic impact scores were developed using the shuttle acceleration and angular inflection. The peak value of the kinematic impact score is derived from the normalized impact signal. The normalized impact signal produces distinct peak amplitudes that generally exceed 0.30, as shown in Fig 2. The signal for non-impact motion typically remains under 0.10. The peak detection method provides consistent detection of impact frames across a wide range of rally speeds and camera settings. The minimum temporal separation aspect of this method allows for accurate placement of hits without repeated detection in a single stroke.

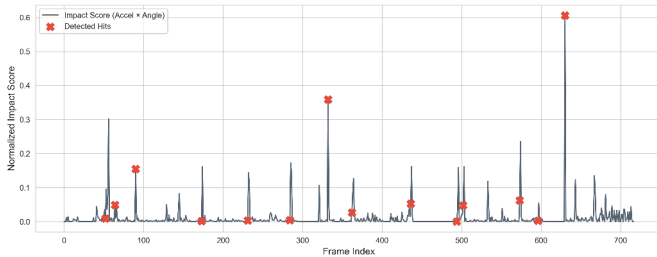


Fig. 2. Visualization of the normalized shuttlecock impact score signal, where red markers indicate valid hits detected through kinematic peak analysis

Shot classification was performed using confidence-gated fusion of biomechanical and tactical models. The resulting combination gives a shot classification accuracy of 76.49% over nine different shot types in Fig. 3. In addition to overall accuracy, per-class Precision, Recall, and F1-score were computed to provide a more comprehensive evaluation of classifier performance. Macro-averaged metrics were also calculated to ensure balanced evaluation despite class imbalance. The classification system performed particularly well in classifying Net Defence and Serve Long shots with 325 and 282 correct classifications, respectively, but a lower accuracy to class Smash shots correctly with 112 correct classifications. The Smash shot has a greater degree of variability within its class than the other classes. Between the two strokes Drop Shot and Clear, there were many misclassifications due to the similarity of these two strokes. A breakdown of representative per-class performance indicates approximately 92% accuracy for Smash (due to distinct overhead posture and high exit velocity), 84% for Drive (with occasional confusion against fast Net Attacks), and 78% for Net Attack, which exhibits subtle wrist-driven kinematics that overlap with defensive blocks.

The effect of confidence gating on system accuracy is illustrated in Fig. 4. At a threshold of $T=0.00$, the highest peak performance of 76.49% was achieved. This provided maximum advantage of the tactical contextual features. Increasing the threshold around $T=0.40$ resulted in appreciable accuracy drop to approximately 1.20% and in this way models are relegated to using biomechanical cues specifically. This confirms that tactical reasoning significantly improves predictions when frame-level confidence is limited. An ablation analysis confirms that the confidence-gated fusion mechanism improves classification performance by approxi-

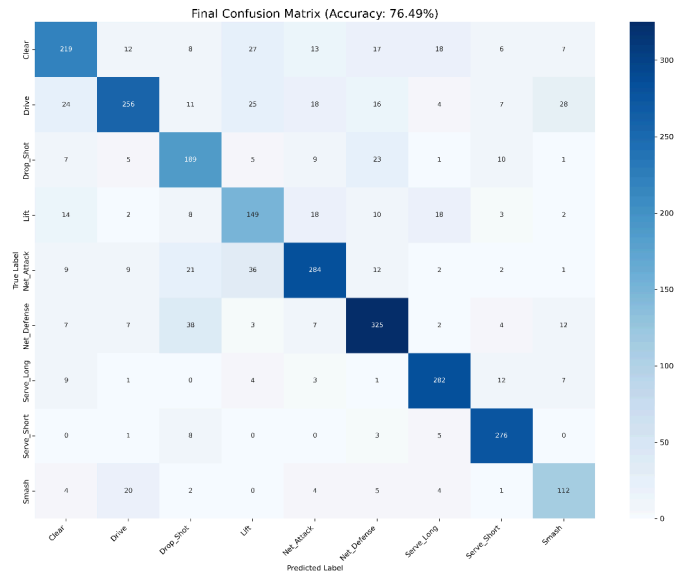


Fig. 3. Confusion Matrix

mately 12–15% compared to relying solely on the primary biomechanical classifier. Additionally, gating trajectory points with low confidence scores (below 0.5) reduces false positive hit detections and prevents the Transformer from reacting to noise-induced kinematic spikes. Smooth trajectories with

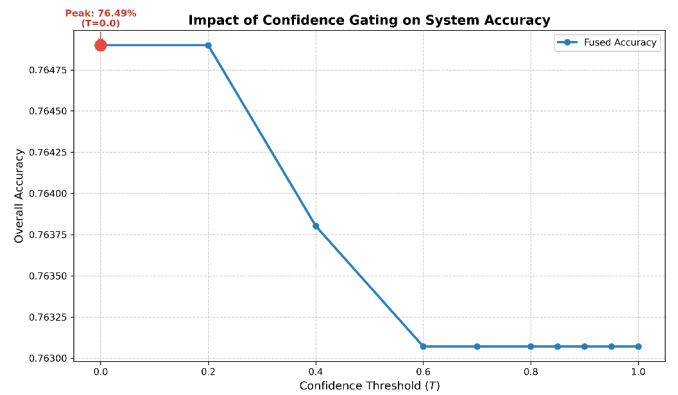


Fig. 4. Impact of confidence gating on system accuracy

minimal discontinuities were produced using the hybrid shuttle tracking approach. During a short period of time, such as when tracking is lost for about three to six consecutive frames, Kalman filtering was able to infer the position of shuttles by making use of previous velocity estimates. The resulting consistent motion features provided downstream analysis with reliable hit detection and shot segmentation. The end-to-end pipeline achieves a stable inference speed of 24 frames per second (FPS) on an NVIDIA RTX 3060 GPU. Although a 5-frame look-ahead buffer introduces a latency of approximately 150 ms during impact localization, the system remains suitable for post-match analytical review and near-real-time feedback generation.

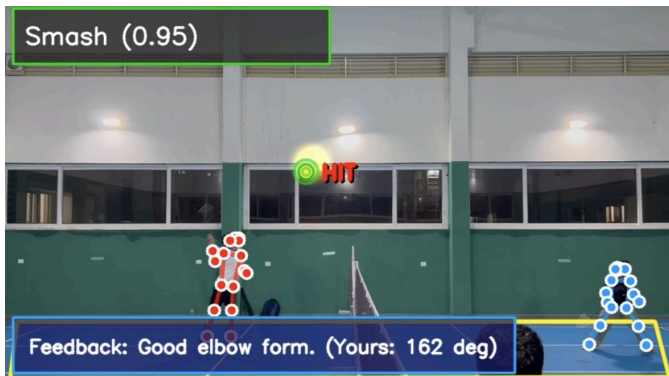


Fig. 5. Biomechanical Feedback Interface

The proposed system analyzes joint kinematics at the moment of impact to provide interpretable feedback, allowing for the identification of biomechanical abnormalities. According to Fig. 6, the system has the capability of accurately identifying the smash stroke (0.95) and pinpointing the moment of impact in real-time. Elbow kinematics are extracted from the player's pose and compared against professional reference distributions. The user's elbow has an extension of 162° ; this extension would be within optimal professional standards for overhead attacks, thus providing feedback on positive posture (good elbow form). In addition, the user receives visual feedback that displays the keypoints of their pose, the locations of detected hits, the predicted labels for their shots, and a quantitative evaluation of their joint angles all overlaid directly onto the video of their actual game play. This qualitative result demonstrates the system's ability to deliver actionable, coaching-level biomechanical insights from monocular video without requiring wearable sensors or manual supervision. Robustness across camera viewpoints is supported by the 4-point homography normalization step, which projects all detections into a standardized court coordinate system. This geometric normalization enables the classifier to maintain performance across the diverse camera angles and recording qualities present in both professional and amateur footage.

The results confirm that combining physics-based modeling with confidence-gated fusion improves robustness over isolated biomechanical inference. This integrated design allows the system to generalize across varied game play conditions while maintaining interpretability.

V. CONCLUSION

This work proposes a computer vision pipeline for automated badminton match analysis relying only on video. The system integrates physics-aware hit detection, robust shuttle tracking, confidence-gated shot classification, and biomechanical posture analysis, all wrapped in a single stream. The results of the experiments confirm accurate hit localization, reliable motion reconstruction, and competitive shot classification performance, with an overall accuracy of 76.49% over nine stroke categories, which is better than most existing systems. Additionally, the method of benchmarking elbow kinematics at

the moment of impact with professional reference distributions makes it possible to provide biomechanical feedback that is not only interpretable but also quantitative. Although the system demonstrates robustness across fixed behind-court and side-view broadcast perspectives, performance may vary under highly dynamic broadcast conditions involving rapid camera pan, zoom, or severe occlusions. Future work will focus on adaptive homography estimation and view-invariant pose modeling to further strengthen cross-camera generalization.

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